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A Synthetic Defect Image Generation Method for Enhanced Industrial Defect Detection Based on Inpainting Diffusion and Context-aware Mask Generation

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Abstract
Automated visual inspection (AVI) systems increasingly rely on deep learning-based object detection to deliver fast and consistent surface defect identification. However, the scarcity of defect image data and the shift in visual feature distributions caused by product variations remain significant challenges that hinder the practical deployment of deep learning systems. To address these challenges, we propose a synthetic defect generation pipeline based on an inpainting diffusion model. We introduce a Context-Aware Mask Generation (CAMG) module to ensure conforming mask generation required for realistic defect image synthesis. The proposed CAMG module employs a Self-DIstillation with NO labels (DINO) vision foundation model to automatically learn defect placement rules on product images. By synthesizing defects onto the non-defective images of new products via an inpainting approach, our method effectively bridges the distributional gap caused by product variations, enhancing robustness in New Product Introduction (NPI) scenarios. We validate the effectiveness of our method on footwear and printed circuit board (PCB) datasets. The experimental results demonstrate that incorporating the synthetic defect images consistently improves the mean average precision (mAP), precision, and recall performances of various defect detection models, with the most notable gains observed in the challenging NPI scenario.

Keywords Synthetic Defect Image Generation · Defect Detection · Inpainting Diffusion · Automated Visual Inspection

Introduction

Automated visual inspection (AVI) systems based on deep learning object detection have become a core component of intelligent manufacturing, replacing the conventional manual visual inspection process. Recent studies have demonstrated promising detection performances in diverse industrial domains such as semiconductors (Shinde et al., 2022), steel (Zhang et al., 2024), railroads (Ni et al., 2022), and agriculture (Lu et al., 2025). However, their practical adoption still faces significant challenges in manufacturing sites. Specifically, manufacturers struggle with long lead times for data acquisition and performance degradation due to the scarcity of defect data and frequent product changes in manufacturing environments. For example, in the footwear manufacturing industry, new product introduction (NPI) occurs frequently (Shen et al., 2020) due to changing demands, and a single production line commonly produces a wide range of products with diverse designs and colorways. Under this dynamic environment, defect detection models continuously face input images of products they have not been trained on. This persistent distribution shift causes a severe degradation in model performance (Lengyel et al., 2023; Yao et al., 2023). Moreover, acquiring real defect data can take several months or more, since defect image data is scarce in manufacturing sites (Garcia-Perez et al., 2025; Li et al., 2023). Consequently, this creates a performance

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lag between the performance degradation and its subsequent recovery as illustrated in Fig. 1, which hinders the widespread adoption of AVI systems across the industry.

To overcome this challenge, we propose a data-centric strategy that enhances defect detection through synthetic defect image generation. The proposed method offers an advantage to manufacturers by accelerating data acquisition and diversifying training datasets while minimizing the dependency on rare real-world defects. Notably, even for a newly introduced product for which no defect data exists, the proposed method requires only a small number of non-defective normal images, which can be rapidly acquired during the prototyping stage (De Giorgio et al., 2023). The generated synthetic images interpolate missing defect patterns within existing product lines, and extrapolate defect features to new products to bridge the distribution shift. As a result, the proposed method mitigates performance degradation during the operational phase in dynamic manufacturing environments.

To evaluate the proposed method under realistic industrial conditions, we conducted extensive evaluations on two distinct real-world industrial datasets, footwear and printed circuit board (PCB). These datasets were specially selected as they represent the target characteristics of our problem setting, multiple product variants within a single category, intricate sub-components, bounding box-level annotations, and highly localized defects. Experimental results confirm that the proposed method consistently outperforms baseline methods, particularly in the challenging NPI scenario. Our approach achieved a significant recall gain of up to 21.5 percentage points on unseen products, highlighting its practical effectiveness in dynamic manufacturing environments. Furthermore, we provided a comprehensive analysis including quantitative evaluation of synthetic image quality and feature space visualization to verify the effectiveness of the proposed generation pipeline.

The main contributions of our research are summarized as follows.

1. We propose a fully automated synthetic defect generation pipeline that integrates Context-Aware Mask Generation (CAMG) with a fine-tuned diffusion model, enabling the data-driven synthesis of contextually valid anomalies without manual intervention.
2. We demonstrate the practical utility of our approach in real-world defect detection datasets, which are characterized by structural complexity, diverse designs and localized defects.
3. Extensive experiments validate that the proposed pipeline consistently enhances the performance of various state-of-the-art defect detection models, effectively mitigating the problems of data scarcity and distribution shift in dynamic industrial environments.

The remainder of this paper is organized as follows. Section Related Work reviews related work on synthetic defect image generation methods including inpainting diffusion that are central to our methodology. Section Methods explains the proposed method in detail. Section Experimental Results presents the experimental results, followed by comparative analysis and discussions. Finally, Section Conclusion concludes the paper and suggests directions for future research.

Related Work

Deep Learning for Industrial Defect Detection

Deep learning-based object detectors, particularly single-stage methods based on You Only Look Once (YOLO) (Redmon et al., 2016) and Detection Transformer (DETR) (Zhao et al., 2024) architectures, have been widely adopted for AVI systems. Their end-to-end architecture is suitable for real-time inference and multi-class defect detection, which is a critical requirement for in-line inspection in manufacturing. However, supervised object detectors rely on large-scale labeled datasets. Insufficient training data and distribution shifts can result in a severe overfitting problem. Data augmentation (Wang et al., 2025) and domain adaptation (Wang et al., 2023) have been widely studied to overcome these challenges. Although domain adaptation methods such as domain adversarial learning (Ganin et al., 2016; Li et al., 2023) enable end-to-end feature alignment between distinct input distributions, they require representative target domain samples during training, which is unavailable when no defect images of the target product exist prior to production. Moreover, these methods often focus on global feature alignment and may inadvertently suppress the fine-grained local features essential for precise defect localization. Consequently, recent paradigms have shifted towards a generative data augmentation approach leveraging large-scale foundation models to enrich the existing dataset.

Synthetic Defect Image Generation

Data augmentation is a widely adopted strategy to address data scarcity in industrial defect detection. Early approaches utilized copy-and-paste methods (Li et al., 2021) and GAN-based methods (Zhang et al., 2021) to synthesize defect images. However, they often fail to generate realistic textures or seamless blends with the background. Recently, latent diffusion models (Rombach et al., 2022) have emerged as a powerful alternative, outperforming previous methods in generating high-fidelity images. In particular, an inpainting diffusion model, a specific type of diffusion model that is trained to fill the mask area seamlessly onto a given input image, showed strong capability to generate local defect

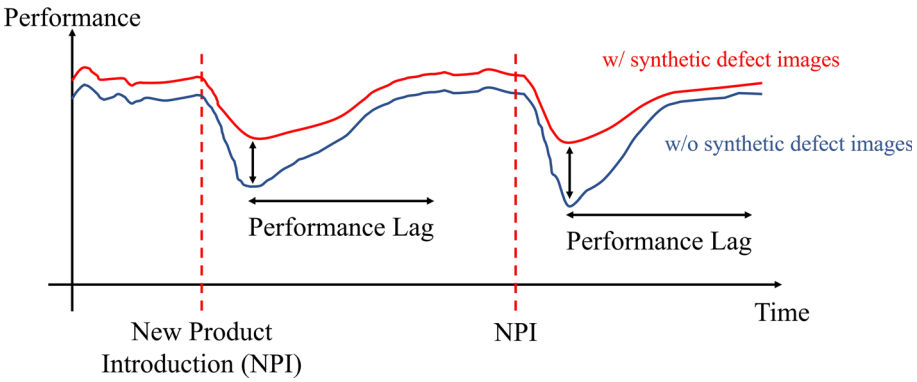


Fig. 1 Performance degradation of defect detection models in new product introduction case

textures while preserving the structure of whole products. Previous studies such as AnoGen (Gui et al., 2025), AnomalyDiffusion (Hu et al., 2024), and DefectFill (Song et al., 2025) demonstrated the generative capability of inpainting diffusion models to enhance defect detection with synthetic defect images on academic benchmarks such as MVTEC-AD (Bergmann et al., 2019) and ViSA (Zou et al., 2022). Recent studies are also extending beyond standard benchmarks to applications on real-world datasets (Chen et al., 2025; Zeng et al., 2025).

Mask Generation for Inpainting Diffusion

In the application of inpainting diffusion models for defect image generation, mask generation serves as a critical step that determines the spatial location and the specific category of the defect to be synthesized. In various industrial domains, defect masks must be carefully located for defect image generation. For instance, in our case study of footwear product defect detection, a 'Loose Thread End' defect must be placed on semantically valid regions such as the stitching area, rather than on incompatible materials like the rubber sole. However, approaches for mask generation in previous studies show several limitations. Reusing defect masks from the training set or using custom-drawn masks can ensure mask validity, but hinder the full automation of the data generation pipeline (Chen et al., 2025; Gui et al., 2025; Song et al., 2025). Foreground sampling (Choi et al., 2025; Rother et al., 2004) distinguishes products from the background, but lacks granularity for precise defect placement within product parts. Finally, while integrating mask generation directly into the generative model can be an effective strategy, such an end-to-end approach presents low interpretability (Hu et al., 2024). To address these problems, we introduce a learnable Context-Aware Mask Generation (CAMG) module that explicitly models the joint distribution of defect location, type, and size conditioned on the input image. Based on vision foundation model features, CAMG enables semantically consistent and contextually valid defect synthesis, particularly for industrial products with complex surface structures.

Parameter-Efficient Fine-tuning for Diffusion Models

While large-scale diffusion models possess strong generative capabilities, they are pre-trained on general-domain datasets and often lack the domain-specific knowledge required to represent industrial defects and complex product textures. Therefore, fine-tuning is essential to bridge this gap and enable the model to capture the distinct visual characteristics of the target products.

To achieve this while preserving the pre-trained capabilities, previous studies have explored various parameter-efficient fine-tuning strategies. In particular, Low-Rank adaptation (LoRA) (Hu et al., 2022) enables layer-wise fine-tuning with a minimal number of learnable parameters. Textual Inversion (Gal et al., 2023) and Dreambooth (Ruiz et al., 2022) introduce learnable conditioning tokens to capture specific concepts. These methods enable fine-tuning of large-scale generative models without the risk of overfitting even with a few-shot supervision. Drawing inspiration from these studies, our pipeline incorporates LoRA and textual inversion to adapt the inpainting diffusion model to the industrial domain.

Methods

The overall framework of the proposed synthetic defect image generation pipeline is illustrated in Fig. 2. A Context-Aware Mask Generation (CAMG) module and the inpainting diffusion model is described in a diagram. Learnable, frozen, and partially learnable parameters are given in colors. I_c, B_c, $\hat{I}_c$ represent cropped images and masks.

Context-Aware Mask Generation (CAMG)

In products composed of multiple sub-components, the types of potential defects vary depending on the specific component. Injecting inappropriate masks during the generation stage results in semantically implausible images, which can ultimately hinder the effective training of defect detectors.

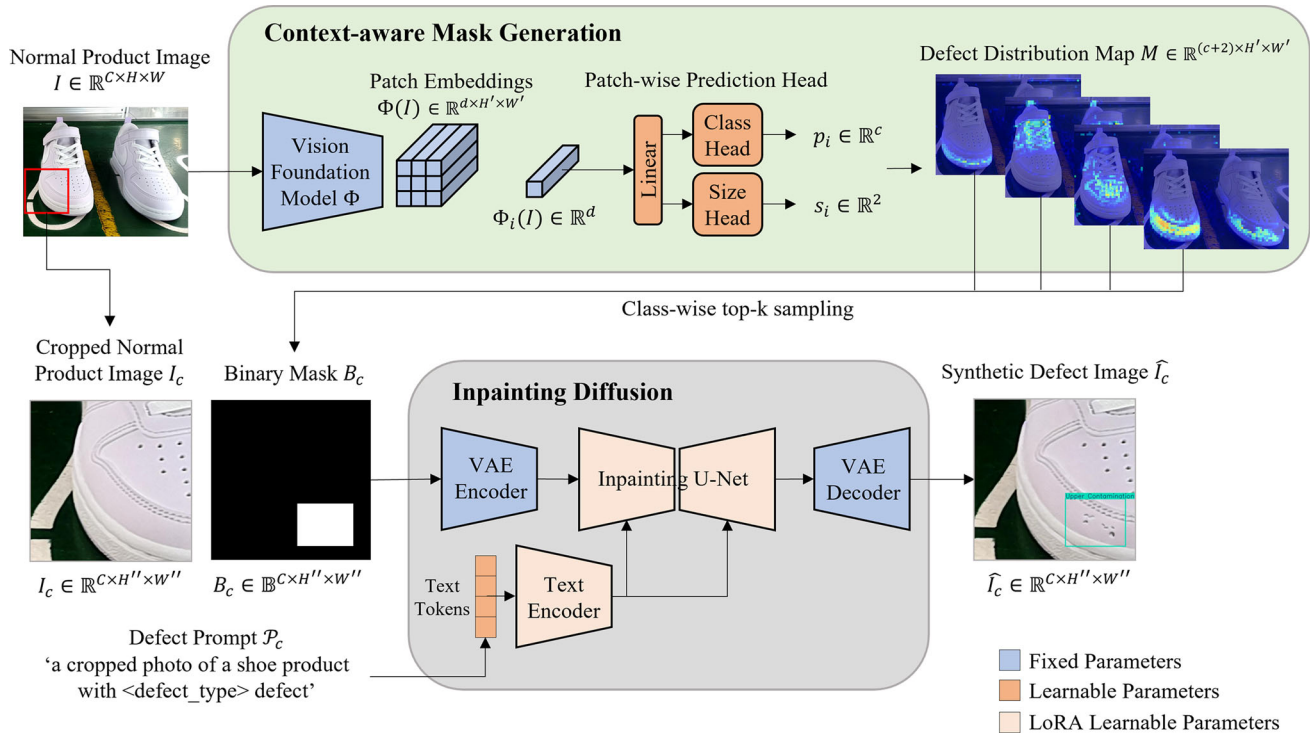


Fig. 2 Framework of the proposed synthetic defect image generation pipeline

Expanding upon established approaches that model object location and appearance as learnable conditional probability distributions for semantically coherent image synthesis (Lee et al., 2018), we propose a Context-aware Mask Generation (CAMG) module. The proposed CAMG module models the defect context as an explicit conditional probability distribution, enabling both fully automated mask generation and disentangled representation of defect context. We define this process as:

$$\mathcal{C}(I) = P(L, T, S|\Phi(I)) \tag{1}$$

where $\mathcal{C}$ is the entire CAMG module, I is an input image, and L, T, S represent the location, type, and size of potential defects respectively. $\Phi(I)$ is a set of patch features obtained from a frozen vision foundation model. The location L is given as prior in $\Phi(I)$, and T, S are the result of two patch-wise prediction heads. Various paradigms of vision foundation models can be used for Φ, such as contrastive language-image grounding (Radford et al., 2021), masking (He et al., 2021), segmentation (Kirillov et al., 2023), and self-distillation (Oquab et al., 2024). The class head and the size head are linear layers that output defect class probabilities p_i and size parameters of bounding boxes $\hat{s}_i = (\hat{w}_i, \hat{h}_i)$. They can be jointly trained by a shared linear layer. We explicitly designed the process to provide box-level guidance rather than pixel-level segmentation masks. Our approach reflects the practical constraints of manufacturing environments, where acquiring pixel-level annotations is labor-intensive and bounding box annotations are predominant (Caruso et al., 2026). In addition, a non-defective patch is classified as a 'no defect' type in the classification head to guide the model to learn to focus exclusively on the foreground area of the product image.

In the training stage, the foundation model is frozen, and two prediction heads are jointly trained with patch features by minimizing a total loss L_{CAMG}, which is a weighted sum of a classification loss L_{cls} and size regression loss L_{box}:

$$L_{cls} = -\sum_{i}\sum_{k=1}^{K+1} y_{i,k} \log p_{i,k} \tag{2}$$

$$L_{box} = ||\log(w_i) - \log(\hat{w}_i)||_2^2 + ||\log(h_i) - \log(\hat{h}_i)||_2^2 \tag{3}$$

$$L_{CAMG} = L_{cls} + \lambda L_{box} \tag{4}$$

Since the defects occur in a local area, using all patch features leads to severe data imbalance problem. Therefore, we use random patch sampling, sampling one positive patch per each defect bounding box instance and three negative patches from randomly selected non-defective areas. This ratio was determined based on the product composition of the dataset. Datasets with extensive backgrounds require a higher negative ratio to capture normal feature variance, whereas tightly

cropped images require a lower ratio. Also, we set $\lambda = 1$, and the size loss is only defined on positive patches.

In the inference stage, all patch features are fed into the heads to produce a dense defect probability map $M \in \mathbb{R}^{(c+2)\times H'\times W'}$, where c is the number of defect classes and H', W' are ($\lfloor H/\text{patch size} \rfloor$, $\lfloor W/\text{patch size} \rfloor$). Then, a binary mask B can be sampled for each defect class using top-k sampling. We use k=5, and this random sampling enhances the diversity of defect images. Then, the image I and mask B are given as input conditions of the following inpainting diffusion model.

Fine-Tuning Inpainting Diffusion Model

Inpainting Diffusion Process

We adopt the Latent Diffusion Model (LDM) (Rombach et al., 2022) as our generative backbone. Specifically, we employ an inpainting diffusion model that synthesizes only masked regions based on the given image. Given an input image I, a binary defect mask B and a defect prompt $\mathcal{P}$, the input image is first encoded into a latent representation using a variational autoencoder (VAE). During the denoising process, the U-Net ϵ_θ predicts the noise component ϵ from the noisy latent x_t at each denoising timestamp t. In addition, after each denoising process, the latent representation corresponding to the unmasked region $(1 - B)$ is overwritten to preserve the background area.

Fine-Tuning Strategy

We propose three main strategies to fine-tune the inpainting diffusion model. First, LoRA layers are added to fine-tune the text encoder and U-Net with a small number of parameters. LoRA layers include two learnable low-rank matrices A and B that approximate the original layer W_0 by

$$W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} B A x \tag{5}$$

where W_0 is the original frozen layer, r is the rank and α is a scaling parameter. These parameters learn the domain-specific visual features, which is not learned in the pre-training stage. Second, new learnable text tokens are added to learn new concepts. Text tokens separate the concepts of different types of defects, and the model can distinguish between different defect types. We assign one special token for each type of defect, and use pre-trained natural language tokens for product names. Third, we adopt two mean squared losses, defect loss $\mathcal{L}_{def}$ and reconstruction loss $\mathcal{L}_{rec}$. Both losses are defined as the differences in the latent space. Defect loss leads the model to inpaint the mask area with defects, while reconstruction loss restricts the model to maintain the structural context of the product. With a randomly selected timestep t, the total loss $\mathcal{L}_{tot}$ is the sum of two losses, and the weight λ controls the balance between two losses. We used $\lambda = 0.1$ in our case study. The losses are defined as follows:

$$L_{def} = \mathbb{E}_{x_t,\epsilon,t}\left[B \odot ||\epsilon - \epsilon_\theta(x_{t,B}, t, \tau(\mathcal{P}))||_2^2\right] \tag{6}$$

$$L_{rec} = \mathbb{E}_{x_t,\epsilon,t}\left[B' \odot ||\epsilon - \epsilon_\theta(x_{t,B'}, t, \tau(\mathcal{P}'))||_2^2\right] \tag{7}$$

$$L_{tot} = L_{def} + \lambda L_{rec} \tag{8}$$

where I is an input image, B and B' are defect masks, $\mathcal{P}$ and $\mathcal{P}'$ are text prompts, and τ is a text encoder. We randomly sample an additional bounding box mask of a non-defect region M' with non-defect prompt $\mathcal{P}'$. Text prompts are set as $\mathcal{P}$='*a photo of a [product name] with [defect type]*' and $\mathcal{P}'$='*a photo of a [product name]*'. $\mathcal{P}$ and $\mathcal{P}'$ are similar except for a special defect token, which allows the defect tokens to focus on the distinctive visual feature of the defects.

Synthetic Defect Image Generation

With the given product image and defect masks, the inpainting diffusion model generates synthetic defect images through an iterative denoising process. In addition, diffusion models are known to occasionally produce undesirable outputs because of the nature of their probabilistic generation process (Choi et al., 2025; Garcia-Perez et al., 2025). The types of undesirable outputs can be divided into three cases: Normal reconstruction, unrealistic defect, and non-conforming mask generation. While the non-conforming mask generation is minimized by the CAMG module, the other two cases can misguide the defect detector. Therefore, we introduce a confidence-based filtering method to achieve a fully automated generation pipeline. We employ a defect detector trained on real images to evaluate the generated images. Then, we rank the synthetic defect instances based on their detection confidence scores and filter out the bottom $k\%$.

Experimental Results

This section explains the dataset used in the case study and the following experimental results. The experiments were conducted using Intel(R) Xeon(R) CPU, NVIDIA A100 GPU, 64GB RAM, Python 3.12, and PyTorch 2.8.0. The inference speed was measured on an edge equipment using Intel(R) Xeon(R) CPU, NVIDIA RTX 3070 GPU, 32GB RAM, Python 3.11 and PyTorch 2.8.0.

Dataset Description

The case study was conducted with two industrial surface defect image datasets. First, the footwear defect image

dataset (Cho et al., 2026) is a defect detection dataset collected from an actual manufacturing line of a footwear manufacturing site, including 2,916 defective images and 476 non-defective images captured from four different viewpoints. Eight different types of products are included in the dataset, and five types of surface defects were annotated in the bounding box format. Second, the printed circuit board (PCB) defect image dataset (Huang & Wei, 2019) includes 8,534 images for training and 1,066 for validation with six types of surface defects, which were collected from 12 different types of PCB products. To simulate the low-data scenario, we used only 10% of the training images randomly. In addition, we added 799 non-defective images to use for the generation process by cropping non-defective areas in the product. Both products share common properties of complex structures and local surface defects. The input images are given in cropped image patches, and the shapes of image patches are 512×512 and 600×600 pixels respectively. Visualizations of defect types are provided in Fig. 3.

Validation of CAMG Module for Learnable Defect Context Modeling

The experimental result in Table 1 shows the performance of various pretrained vision foundation model features for the backbone of the CAMG module in the footwear dataset. An Adam optimizer with the learning rate of 1e-4 was used, and the training process iterated for maximum 10 epochs using batch size 32. The result shows that DINOv2 features outperform the other state-of-the-art foundation models with a large margin. We attribute this superiority to the self-distillation training objective of DINO, which provides part-level semantic understanding. In contrast, MAE focuses on pixel-level reconstruction, resulting in missing high-level semantics. SigLIP and SAM prioritize global image-text alignment and geometric segmentation, which lead to lower discriminative capabilities. In addition, leveraging the emergent properties of DINO features provides robustness under stylistic variations, such as colors, materials, and structures (Caron et al., 2021).

Defect probability maps produced by CAMG module using a DINOv2 backbone are presented class-by-class in Fig. 4. In case of footwear products, two results are shown to demonstrate the robustness over diverse viewpoints. The figure shows how CAMG module can provide contextually valid defect masks. For example, in the footwear product image, 'High Cement' and 'Bottom Contamination' defects only occur around the rubber sole area, and 'Upper Contamination' and 'Loose Thread End' defects occur on the upper area of the product. Note that 'Wrinkle' defect is rarely present in the defect probability map, since the wrinkle defect is mainly observed on the rear side of the product. Furthermore, the probability map correctly divides the product area and background area, which is related to the object-awareness of DINO features (Oquab et al., 2024). In the PCB product image, while 'Spur' defect occurs across the product, 'Missing Hole' only occurs on the soldering area. These probability maps show that the CAMG module effectively models the context of defects without using any explicit prior.

Synthetic Defect Image Generation

Generation Process

We fine-tuned a pre-trained inpainting diffusion model (Rombach et al., 2022). LoRA layers were added to the attention projection layers and output projection layers of U-Net and text encoder with parameters of $r = 8, \alpha = 16$. A total of 46 million learnable parameters were fine-tuned for 10 epochs using a batch size of 32. All modules were trained with an Adam optimizer, with the learning rate of 1e-4 for U-Net and 5e-5 for text encoder. These hyperparameters were selected following previous studies (Song et al., 2025). The size of negative masks were randomly selected between the first and third quartile of the bounding box size distribution of each dataset. The fine-tuning process took 7 to 9 hours. In the generation stage, we randomly selected one defect patch for each defect type using top-k sampling. Defect images were generated by iterating 50 denoising steps. A negative prompt *'blurry, low quality, deformed, ugly, watermark, signature, text, logo'* was given together to prevent the model from generating unwanted artifacts. The generation process took 4 seconds per image under the given setting. A total of 1,931 and 829 defect images were generated in the footwear and PCB dataset respectively.

Generation Quality Evaluation

We employ four quality metrics for the evaluation of generated images, Foreground Structural Similarity Index Measure (Foreground SSIM), Fréchet Inception Distance (FID) (Heusel et al., 2018), Learned Perceptual Image Patch Similarity (LPIPS) (Zhang et al., 2018) and pseudo recall. Foreground SSIM computes the structural similarity between the original and generated images within the mask region, directly assessing how seamlessly the generated defect blends with its background. A higher SSIM value indicates that the generative model successfully preserves and reflects the structural context. FID and LPIPS are utilized to evaluate the visual fidelity and diversity of the generated samples at the distribution level. The FID score is low when the distribution of the synthetic images is closer to that of the real defect images. Conversely, LPIPS measures the perceptual variance of the generated images. While a higher LPIPS score is conventionally favored in general image synthesis to demonstrate high diversity, an excessively

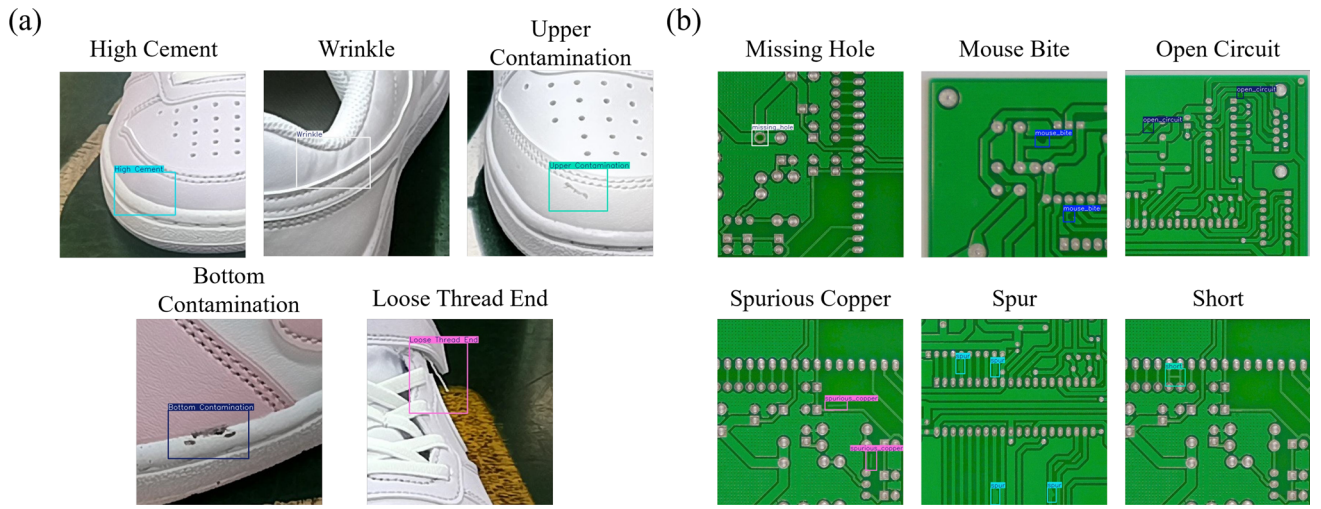


Fig. 3 Defect types in two industrial defect detection datasets. (a) Footwear defect image dataset. (b) PCB defect image dataset

Table 1 Experimental results of patch-wise classification and size regression with vision foundation models in the CAMG module

Foundation Model Φ	Feature Dim. d	Classification Loss L_{cls}	Size Loss L_{box}
DINOv2 (Oquab et al., 2024)	768	**0.293**	**0.007**
SigLIP (Zhai et al., 2023)	1152	0.431	0.013
MAE (He et al., 2021)	768	0.780	0.009
SAM (Kirillov et al., 2023)	256	0.493	0.008

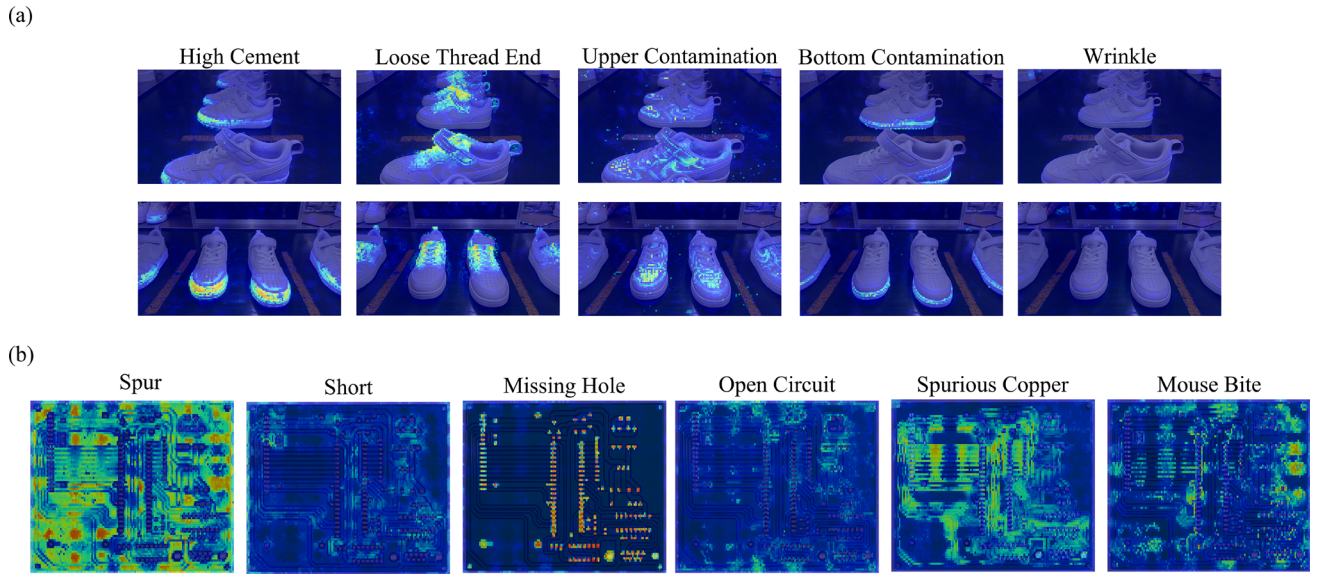


Fig. 4 Defect probability maps produced on non-defective product images. (a) Footwear dataset. (b) PCB dataset

high LPIPS score often indicates the generation of unrealistic over-exaggerated defects. Therefore, we prefer an LPIPS score that closely approximates the intrinsic perceptual variance of the real dataset. Pseudo recall measures the degree of alignment between the synthetic defect images and the behavior of a reference detector trained exclusively on real data. Specifically, it is computed as the recall of the reference detector on the generated images, which is motivated by the TRTS (train on real, test on synthetic) evaluation paradigm (Esteban et al., 2017). A substantially low pseudo recall suggests that the generated defects deviate significantly from learnable defect patterns, which introduce noise into the training process.

The evaluation results are shown in Table 2. *Random* in the table indicates random sampling for mask generation, and *CutPaste* indicates crop and paste method (Li et al., 2021). *Foreground* indicates GrabCut (Rother et al., 2004) foreground extraction algorithm, which is suggested in AnoGen (Gui et al., 2025) pipeline. *DeepFillv2* is a GAN-based inpainting method (Yu et al., 2019) implemented for the comparison. The results clearly show how incorporating the CAMG and diffusion model improves the quality of synthetic defect images. Our method outperformed the baseline methods, with the highest foreground SSIM, the lowest FID, and the lowest LPIPS difference compared to the real defect images. Notably, both random and foreground sampling mask generation methods paired with diffusion model yield competitive FID and LPIPS scores, yet their pseudo recall remains marginally lower than that of our method across both datasets. This discrepancy suggests that the synthesized defects fail to align with the defect patterns when defect masks are not positioned in contextually appropriate regions. Foreground SSIM scores are measured only within our mask generation method to fairly compare the inpainting capabilities of generation methods.

For the qualitative evaluation, sample synthetic defect images of all defect types in each dataset are shown in Fig. 5. Results in the figure demonstrates that the synthetic defect images exhibit high fidelity, while differentiating the distinct visual features of each defect type using text prompts. In addition, a comparison between GAN-based method is provided in Fig. 6. Red box regions in the figure reveals that the GAN-based method has limitations in fully reproducing the texture of the real image data. Furthermore, the figure also demonstrates that despite being trained to generate defect data, the model lacks sufficient representational capability to synthesize localized defects.

Failure Cases and Confidence-based Filtering

Three types of failure cases are illustrated in Fig. 7. Original images and three types of failure cases, (a) normal reconstruction, (b) unrealistic defects, and (c) non-conforming masks are shown in the figure. To exclude low-quality images such as those in the figure, we used a filtering ratio $k = 25\%$ based on experiments to balance between the diversity and quality of synthetic defect images. A YOLO11 model trained with real images was used for filtering.

Defect Detection with Synthetic Defect Images

To evaluate the efficacy of synthetic defect images, we measured the change in the performance on the defect detection models when these images were included in the training dataset. Three recent versions of You Only Look Once (YOLO) network architectures (Redmon et al., 2016), YOLOv8, YOLOv9, and YOLO11 (Jocher & Qiu, 2024; Jocher et al., 2023; Wang et al., 2024), and two Detection Transformer (DETR) architectures (Zhao et al., 2024) with Vision Transformer (ViT) backbone and Resnet50 backbone were tested in the experiment. These object detection models are widely employed for industrial defect detection (Hussain, 2023), showing cutting-edge performance at real-time inference speeds. Experiments were conducted under two manufacturing scenarios: The In-Distribution (ID) case and the New Product Introduction (NPI) case. Fig. 8 conceptually illustrates the data split and generation flow for both the ID and NPI scenarios. To construct the two scenarios from a shared dataset, a subset of products was first randomly designated as unseen products and withheld from the training by a product-wise split. The remaining products formed the common training set used in both scenarios. In the ID case, the unseen products are excluded, and we used 85% of randomly selected retained images for training and the other 15% for validation. In the NPI case, the training set is equivalent, while the validation set is replaced with images from the withheld unseen products. Specifically, 6 out of 8 products for footwear and 9 out of 12 products for PCB were used for training, and the remaining products were used for validation. Note that a small set of normal images for unseen products is assumed to be accessible in advance, which reflects a practical NPI scenario where sample product images can be collected before defects are observed. An IoU threshold of 0.25 and a confidence threshold of 0.25 were applied for precision and recall measure.

All object detectors were trained under same setting for fair comparison. The scale parameter is 'small' for YOLO-based models, considering the number of patch-wise inferences required in the deployment stage (Akyon et al., 2022; Cho et al., 2026). Models were trained for 100 epochs using an AdamW optimizer with 1e-4 learning rate, and with batch size 8. In addition, HSV augmentations, random flip, random rotation were applied on the training images for better generalization. Mean average precision (mAP50), precision, and recall metrics presented on this section are averaged across all classes. Additional class-wise experimental results are reported in Appendix Appendix A. All performance gains are expressed in percentage points (pp).

In-Distribution Case

The experimental results in the ID case are shown in Table 3. The results show that object detection models clearly benefit from using the synthetic defect images. All models achieved higher mean average precision at IoU 0.5 (mAP50), precision, and recall scores when trained with synthetic defect images in both footwear and PCB dataset. The improvements of mAP50 ranged from 0.8 pp to 7.3 pp in the footwear dataset, and 5.7 pp to 10.6 pp in the PCB dataset.

Table 2 Synthetic defect image generation quality metrics

Dataset	Methods	Foreground SSIM(↑)	FID(↓)	LPIPS	Pseudo Recall(%)
Footwear	Random+CutPaste (Li et al., 2021)	-	68.60	0.741	25.7
	Random+Diffusion	-	26.77	0.735	24.0
	Foreground+Diffusion (Gui et al., 2025)	-	27.15	0.712	37.2
	CAMG+CutPaste	0.501	68.59	0.724	31.8
	CAMG+DeepFillv2 (Yu et al., 2019)	0.563	28.01	**0.709**	26.4
	CAMG+Diffusion(Ours)	**0.739**	**26.27**	**0.709**	**63.4**
	Real Data	-	-	0.687	68.9
PCB	Random+CutPaste	-	201.35	0.515	38.0
	Random+Diffusion	-	45.06	**0.488**	36.9
	Foreground+Diffusion	-	43.30	0.491	30.7
	CAMG+CutPaste	0.615	50.15	0.502	46.1
	CAMG+DeepFillv2	0.766	43.54	0.494	13.4
	CAMG+Diffusion(Ours)	**0.885**	**42.72**	0.493	**75.7**
	Real Data	-	-	0.479	61.2

Table 3 Experimental results in the In-distribution case setup. R: Real data, R+S: Real and synthetic data

Dataset	Model	Data	mAP50 (%)	Precision (%)	Recall (%)	GFLOPS	FPS
Footwear	YOLOv8	R	55.3	79.3	71.3	28.4	116
		R+S	56.1	83.9	72.1		
	YOLOv9	R	56.2	78.2	69.2	26.7	128
		R+S	57.2	82.9	71.9		
	YOLO11	R	47.2	74.5	68.9	21.5	131
		R+S	51.0	82.2	71.5		
	RT-DETR-ViT-L	R	54.7	73.4	68.2	103.5	45
		R+S	**60.5**	**79.2**	**78.5**		
	RT-DETR-Resnet50	R	48.9	66.0	64.5	125.6	44
		R+S	56.2	75.1	73.6		
PCB	YOLOv8	R	54.8	65.0	63.2	28.4	116
		R+S	60.5	65.7	71.4		
	YOLOv9	R	62.0	69.8	74.2	26.7	128
		R+S	**70.6**	**71.5**	**81.1**		
	YOLO11	R	53.5	62.3	61.2	21.5	131
		R+S	61.9	69.4	68.0		
	RT-DETR-ViT-L	R	52.4	61.7	66.1	103.5	45
		R+S	63.0	70.4	77.5		
	RT-DETR-Resnet50	R	55.6	56.7	61.0	125.6	44
		R+S	65.0	66.7	71.7		

Precision improvements made by the synthetic dataset were up to 9.9 pp and 10.0 pp respectively, and recall improvements were up to 10.3 pp and 11.4 pp. The best results in each dataset are marked in bold. RT-DETR with ViT-L backbone achieved the highest performance in the footwear dataset, while YOLOv9 showed the best results in the PCB dataset. Regardless of the specific architecture, consistent performance gains across all models confirm that using synthetic defect images is a robust strategy for enhancing defect detection accuracy. These results demonstrate that the synergy between optimal model selection and synthetic data generation is the key to achieving state-of-the-art performance in industrial defect detection.

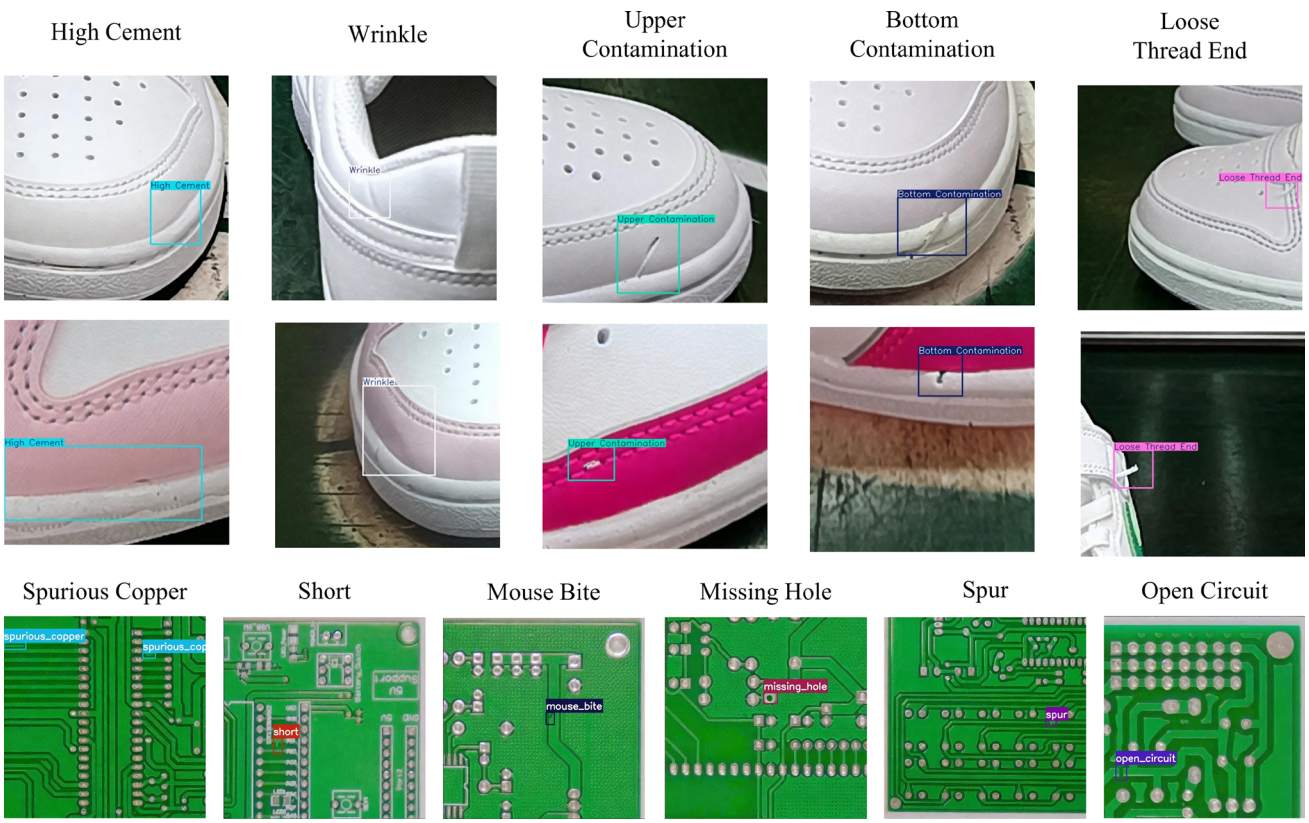


Fig. 5 Synthetic defect images

Table 4 Comparison of defect detection performance between synthetic defect image generation methods. R: Real data, R+S: Real and synthetic data

Dataset	Data	Method	mAP50 (%)	Precision (%)	Recall (%)
Footwear	R	-	47.2	74.5	68.9
	R+S	Random+Diffusion	47.0	72.6	63.5
		Foreground+Diffusion (Gui et al., 2025)	42.1	67.7	62.8
		CAMG+DeepFillv2 (Yu et al., 2019)	46.1	65.8	64.5
		Ours	**51.0**	**82.2**	**71.5**
PCB	R	-	53.5	62.3	61.2
	R+S	Random+Diffusion	50.6	58.0	60.2
		Foreground+Diffusion	49.2	57.9	56.2
		CAMG+DeepFillv2	56.1	64.7	65.1
		Ours	**61.9**	**69.4**	**68.0**

Table 4 compares the defect detection performance of the YOLO11 model when trained on real images combined with synthetic image sets generated by different methods. The experimental results demonstrate that our proposed pipeline achieved the highest performance improvements across both datasets, recording significant mAP gains of 3.8 pp and 8.4 pp over real-only baselines in the footwear and PCB datasets, respectively. Conversely, incorporating synthetic defect images from the other baseline methods actually degraded the detection performance in the footwear dataset, leading to lower overall metrics. These differences highlight the importance of two factors that are required to effectively utilize synthetic data for industrial datasets with complex surfaces, a generative model capable of producing high-fidelity textures, and a mask generation process that comprehends domain context to accurately specify defect characteristics.

Table 5 presents the performance variance of the YOLO11 model in the ID case across different synthetic defect image



Fig. 6 Synthetic defect images generated by GAN-based and diffusion-based inpainting methods

Table 5 Experimental results in the In-distribution case setup with different synthetic image filtering ratio. R: Real data, R+S: Real and synthetic data

Dataset	Data	Filtering Ratio k (%)	# of Images	mAP50 (%)	Precision (%)	Recall (%)
Footwear	R	-	2,479	47.2	74.5	68.9
	R+S	0	4,410	47.9	**84.8**	67.0
	R+S	25	3,927	**51.0**	82.2	71.5
	R+S	50	3,444	50.1	81.1	**72.9**
PCB	R	-	853	53.5	62.3	61.2
	R+S	0	1,682	**63.0**	69.1	63.8
	R+S	25	1,474	61.9	**69.4**	68.0
	R+S	50	1,268	62.3	68.3	**70.1**

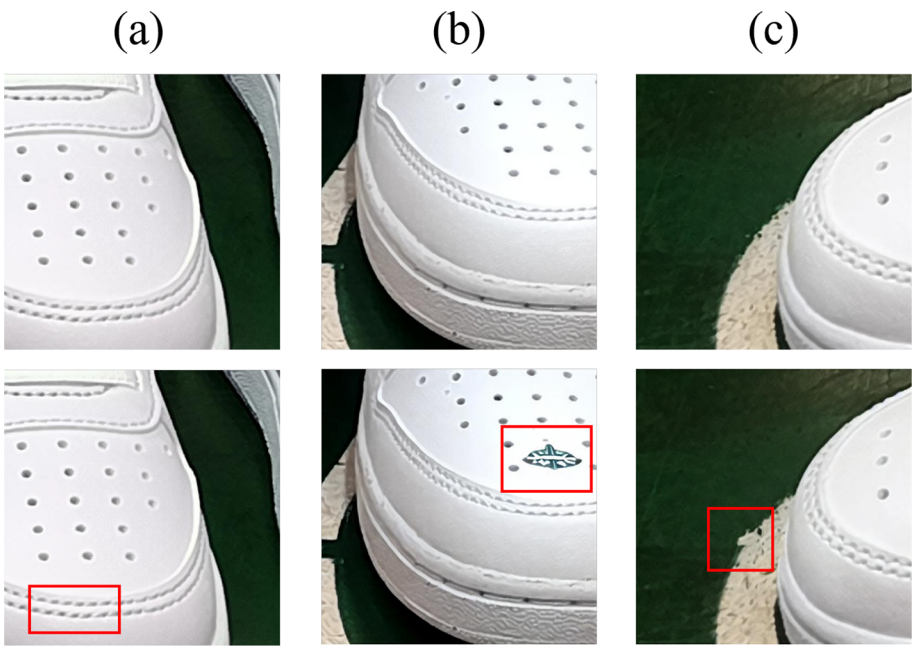


Fig. 7 Three types of failure cases. (a) Normal reconstruction. (b) Unrealistic defect. (c) Non-conforming mask generation

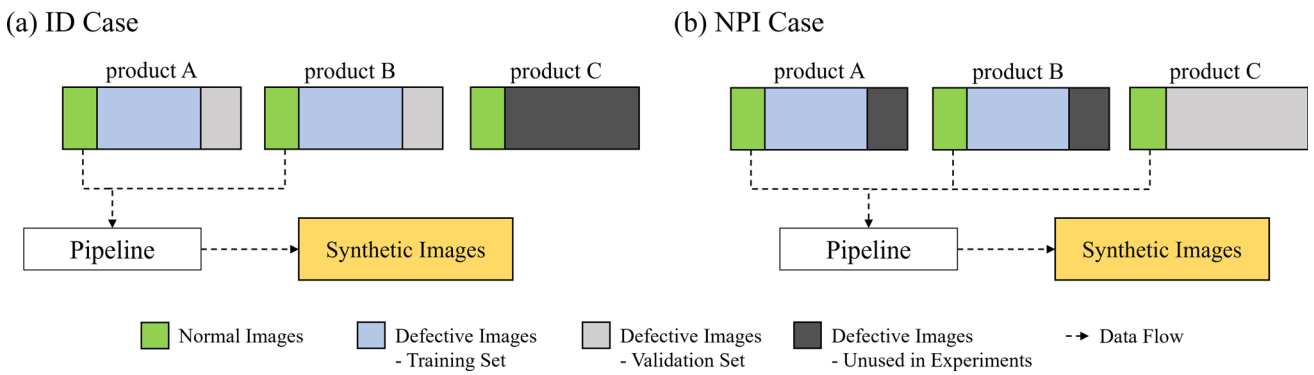


Fig. 8 Conceptual illustration of two experimental settings

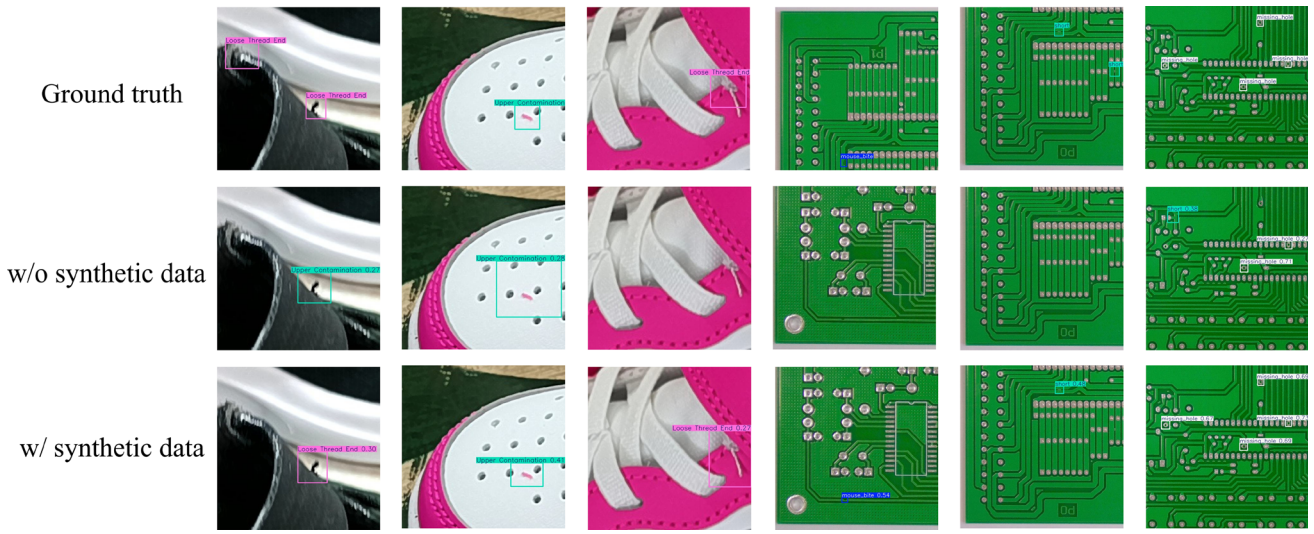


Fig. 9 Comparison of defect detection results between YOLO11 models in NPI case trained with and without synthetic defect images

Table 6 Experimental results in the NPI case setup. R: Real data, R+S: Real and synthetic data

Dataset	Model	Data	mAP50 (%)	Precision (%)	Recall (%)	GFLOPS	FPS
Footwear	YOLOv8	R	37.3	58.0	45.2	28.4	116
		R+S	40.5	66.0	55.6		
	YOLOv9	R	37.4	59.8	45.8	26.7	128
		R+S	43.1	67.7	58.7		
	YOLO11	R	36.6	58.1	41.0	21.5	131
		R+S	42.0	71.0	**59.3**		
	RT-DETR-ViT-L	R	42.4	65.4	48.4	103.5	45
		R+S	**46.0**	**73.9**	58.0		
PCB	YOLOv8	R	40.0	58.9	58.1	28.4	116
		R+S	56.4	64.7	69.3		
	YOLOv9	R	38.4	53.0	48.5	26.7	128
		R+S	57.6	66.2	70.0		
	YOLO11	R	34.2	52.3	52.3	21.5	131
		R+S	54.3	66.0	65.8		
	RT-DETR-ViT-L	R	45.6	58.1	65.0	103.5	45
		R+S	**61.1**	**69.0**	**75.6**		

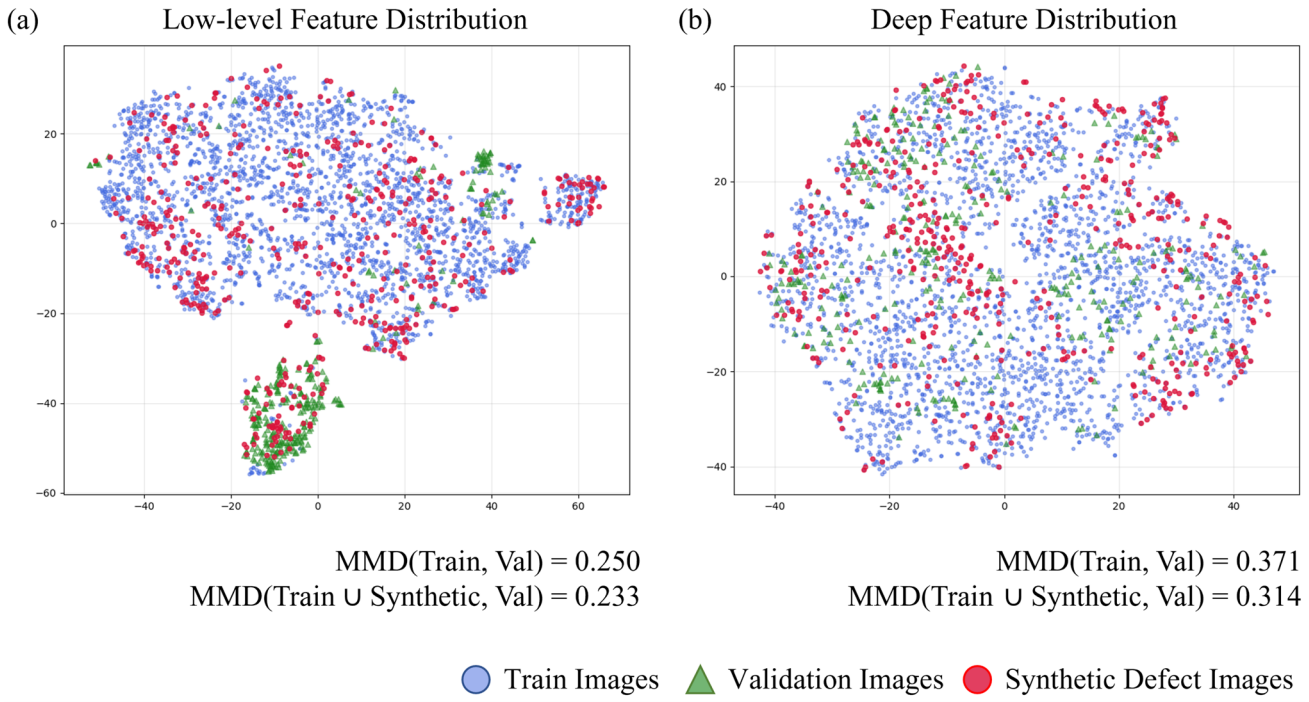


Fig. 10 t-SNE visualization of feature distributions extracted from YOLO11 backbone. (a) Low-level feature distribution. (b) Deep feature distribution

filtering ratio k. A trade-off between precision and recall is observable in both datasets. A lower k tends to increase precision but decrease recall, whereas applying higher k improves recall at the expense of precision. In addition, the highest mAP50 in the PCB dataset was achieved without any filtering. This result is related to the difficulty of defect image generation. Since the PCB dataset features relatively monotonic defect patterns, the generative model inherently produces reliable synthetic images, thereby reducing the necessity of the filtering process.

Table 7 Detailed Experimental Results on Footwear Dataset. HC-High Cement, UC-Upper Contamination, BC-Bottom Contamination, LTE-Loose Thread End, WR-Wrinkle

Setup	Model	Data	mAP50 (%)	Class-wise mAP50 (%)				
				HC	UC	BC	LTE	WR
ID	YOLOv8	R	55.3	60.1	59.6	50.0	53.8	52.9
		R+S	56.1	63.4	65.7	55.3	54.5	41.5
	YOLOv9	R	56.2	60.2	50.8	57.9	56.5	55.7
		R+S	57.2	58.1	54.8	52.8	57.1	63.0
	YOLO11	R	47.2	54.9	52.4	45.7	42.0	40.9
		R+S	51.0	60.7	53.4	53.3	47.7	39.9
	RT-DETR-ViT-L	R	54.7	46.9	32.2	62.6	57.1	69.8
		R+S	60.5	52.6	50.7	65.4	63.4	70.6
	RT-DETR-Resnet50	R	48.9	38.6	52.5	49.0	46.6	57.7
		R+S	56.2	45.3	56.9	59.5	58.8	60.5
NPI	YOLOv8	R	37.3	30.6	35.2	52.0	29.5	39.0
		R+S	42.0	36.9	36.9	53.4	32.8	50.0
	YOLOv9	R	37.4	37.8	30.4	50.8	27.2	41.0
		R+S	43.1	38.7	34.1	55.5	32.7	54.7
	YOLO11	R	36.6	23.7	33.4	45.6	35.1	45.3
		R+S	42.0	31.3	36.8	48.0	38.4	55.6
	RT-DETR-ViT-L	R	42.4	35.5	30.7	59.9	53.7	32.0
		R+S	46.0	38.8	36.0	61.8	58.0	35.5

NPI Case

In the NPI case setup, the defect detection models were trained with additional synthetic defect images that are made using the non-defective product images of unseen products. Experimental results are reported in Table 6.

As presented in Table 6, the baseline models suffer from severe performance degradation in the NPI scenario due to the distribution shift between existing and unseen products compared to the performance metrics in Table 3. The results show that incorporating the synthetic defect images significantly mitigates this performance lag across all models, resulting in higher mAP score by up to 5.7 percentage points in the footwear dataset and 20.1 percentage points in the PCB dataset. Improvements in precision and recall scores were up to 12.9 pp, 18.3 pp in the footwear dataset, and 13.7 pp, 21.5 pp in the PCB dataset. The overall performance gain was higher in the PCB dataset. This difference comes from different structural characteristics of the dataset. PCB images feature more rigid structure with consistent spatial alignment, which allows more effective feature space interpolation between different products. Lightweight models trained with synthetic data achieved comparable or superior performance to the computationally expensive models trained solely on real images. This indicates that a data-centric approach is particularly effective than a model-centric approach in NPI scenarios.

Discussion

In this section, we provide a qualitative analysis and a feature-level visualization to explore how the proposed synthetic data generation enhances defect detection performance. Fig. 9 shows different inference results of YOLO11 models trained with and without synthetic defect images. The model trained without synthetic defect image does not capture the defects on unseen products. However, the model trained with synthetic data successfully captures the defect and correctly classifies the defect types.

Fig. 10 reveals how the synthetic data contributes to the significant performance improvements in the NPI case. The figure illustrates the feature distributions of footwear defect images extracted from the backbone of the YOLO11s model trained on real images. Low-level features in (a) and deep features in (b) correspond to the outputs of the first and the last convolutional blocks, respectively. In the low-level feature space, a distinct separation between the train and validation image clusters reflects the visual appearance differences between products. This discrepancy indicates a distribution shift, which is a primary factor contributing to the performance degradation. Conversely, in the deep feature space, the training and validation image features are mixed. This overlap suggests that the intrinsic characteristics required for defect detection remain consistent. The synthetic defect images span across both the training and validation clusters in the low-level space, indicating that the generated images capture the visual traits of the new product. In addition,

Table 8 Detailed Experimental Results on PCB Dataset. MB-Mouse Bite, SP-Spur, MH-Missing Hole, SH-Short, OC-Open Circuit, SC-Spurious Copper

Setup	Model	Data	mAP50 (%)	Class-wise mAP50 (%)					
				MB	SP	MH	SH	OC	SC
ID	YOLOv8	R	54.8	54.6	44.7	53.0	58.1	61.6	57.0
		R+S	56.1	57.1	54.9	63.4	70.6	58.7	58.3
	YOLOv9	R	56.2	63.2	58.3	49.4	69.6	66.0	65.5
		R+S	57.2	69.6	66.5	68.0	74.8	68.5	76.3
	YOLO11	R	53.5	57.1	43.1	50.7	56.5	57.9	55.7
		R+S	61.9	59.4	54.5	62.3	68.9	65.3	60.7
	RT-DETR-ViT-L	R	52.4	57.6	55.7	53.4	51.4	45.0	51.2
		R+S	63.0	71.1	64.7	69.3	63.4	53.3	56.4
	RT-DETR-Resnet50	R	55.6	59.2	49.8	59.2	51.5	52.3	61.6
		R+S	65.0	65.7	67.4	67.2	66.6	59.0	64.3
NPI	YOLOv8	R	40.0	49.4	26.2	28.4	50.1	40.0	45.8
		R+S	56.4	55.2	39.7	65.1	60.8	47.8	69.9
	YOLOv9	R	38.4	44.0	23.5	36.2	47.6	32.0	47.2
		R+S	57.6	53.6	44.0	66.8	62.2	49.0	70.2
	YOLO11	R	34.2	45.5	19.7	35.3	39.8	30.2	34.4
		R+S	54.3	56.7	33.0	60.5	65.9	48.7	61.0
	RT-DETR-ViT-L	R	45.6	45.8	53.6	36.7	41.5	43.6	52.4
		R+S	61.1	56.3	68.6	62.9	65.9	51.3	61.6

the synthetic images are uniformly distributed throughout the cluster in the deep feature space, confirming that the defect-related features of the synthetic data align well with the distribution of real defects. These observations explain how the synthetic data acts as a bridge between the existing and new data distributions and enhances the defect detection models. A quantitative analysis using Maximum Mean Discrepancy (MMD) below the figure also confirms that the inclusion of synthetic images bridges the distributional gap between the training and validation sets.

Conclusion

In this paper, we addressed the challenge of defect data scarcity in dynamic manufacturing environments by proposing a fully automated synthetic defect image generation pipeline. The proposed method can be immediately applied to newly introduced products using only normal images, making it particularly suited for manufacturing settings where real defect images are rare and product designs change frequently. Experiments across multiple detector architectures on two real-world industrial datasets confirmed that the synthetic images consistently improve defect detection performance, with the most substantial gains observed in the NPI scenario. These results suggest that automating the defect data generation process is a practical and scalable direction for maintaining robust inspection systems under distribution shift.

The limitations of our study and future research opportunities are as follows. First, the current method is built on an assumption that the defect types are fixed. The generation capability of entirely novel defect types is not studied yet, and future work could explore few-shot or zero-shot learning techniques for CAMG and inpainting diffusion. Second, the current study generates synthetic defect images by determining the defect's type, size, and location. This approach does not reflect other contexts, such as the defect's orientation or severity. Enhancing the controllability of the defect image generation model could further increase the diversity of the dataset and improve its utility in downstream tasks.

Detailed Experimental Results

Table 7 and Table 8 present the experimental results of class-wise mAP50 metrics in two datasets. 'ID' and 'NPI' setup corresponds to the in-distribution and NPI cases. Class names written in abbreviations are provided in the caption of each table.

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manuscript. Hyunbo Cho provided guidance on the conceptual design and manuscript writing.

Declarations

Competing Interests The authors have no relevant financial or non-financial interests to disclose.

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